Daily Logic Drill #3

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Compound Implications · Truth Tables & Model Counting · Exactly-One-True Deductions · Implication Chaining.

Section A Rapid Recognition

1. Determine whether the compound statement $((P \Rightarrow Q) \vee (Q \Rightarrow P))$ is a tautology (always true regardless of the truth values of propositions P and Q).

Ans: True.

2. If the conditional statement $(P \vee Q) \Rightarrow (R \wedge S)$ is false, and R is known to be true, what is the truth value of proposition S?

Ans: False.

3. Given that $P \Rightarrow Q$ and $Q \Rightarrow R$ are both true, but R is false, what are the truth values of P and Q?

Ans: Both false (P = False, Q = False).

4. For three independent propositions P, Q, R, for how many of the 8 possible truth assignments is the implication $(P \wedge Q) \Rightarrow R$ false?

Ans: 1.

5. For three independent propositions P, Q, R, for how many of the 8 possible truth assignments is the implication $(P \vee Q) \Rightarrow R$ false?

Ans: 3.

6. Given propositions P and Q, if $P \Rightarrow Q$ is true and $Q \Rightarrow P$ is false, what is the truth value of the biconditional $P \iff Q$?

Ans: False.

7. True or false: The statement $P \Rightarrow (Q \Rightarrow R)$ is logically equivalent to $(P \wedge Q) \Rightarrow R$.

Ans: True.

8. State the negation of: "If the polynomial $f(x)$ has a root at $x = 1$, then $f(1) = 0$ and $f'(1) = 0$."

Ans: "$f(x)$ has a root at $x = 1$, and ($f(1) \neq 0$ or $f'(1) \neq 0$)."

‘

9. If $P \Rightarrow Q$, $Q \Rightarrow R$, and $R \Rightarrow P$ are all true, what can you deduce about the truth values of P, Q, R?

Ans: They all share the same truth value (either all true or all false).

10. Consider the four statements: (1) $P \vee Q$, (2) $P \Rightarrow R$, (3) $Q \Rightarrow R$, and (4) $\neg R$. Is this set of four statements consistent (can they all be simultaneously true)?

Ans: No (they are inconsistent).

Section B Manipulation Drills

1. Three statements about an integer n are given:

I. n is divisible by 4.

II. n is odd.

III. n is divisible by 2.

Given that **exactly one** of I, II, III is true, which statement is impossible to be the true one? *(after TMUA 2016 Paper 2 Q4)*

Ans: Statement I.

2. Consider three propositions P, Q, R and the implications $S_1 : P \Rightarrow Q$, $S_2 : Q \Rightarrow R$, and $S_3 : R \Rightarrow P$. Given that **exactly two** of S_1, S_2, S_3 are true, can P, Q, R all have the same truth value? *(after TMUA 2020 Paper 2 Q20)*

Ans: No.

3. Three urns A, B, C each have a label:

Urn A: "The prize is in this urn."

Urn B: "The prize is not in Urn A."

Urn C: "The prize is not in this urn."

Given that the prize is in exactly one urn and **exactly one** label is true, which urn contains the prize?

Ans: "Urn C."

4. For three propositions P, Q, R, determine the number of truth assignments (out of 8) for which $((P \Rightarrow Q) \Rightarrow R)$ is true.

Ans: 5.

5. Three suspects X, Y, Z are questioned about a crime committed by exactly one person:

X: "Y did it."

Y: "Z did not do it."

Z: "I did not do it."

Given that the guilty person is the only liar (the two innocent suspects tell the truth), who committed the crime? *(after TMUA 2017 Paper 2 Q2)*

Ans: X

6. Four statements about a positive integer n are given:

S_1: $n > 2$ is prime.

S_2: n is odd.

S_3: n is a multiple of 3.

S_4: $n = 9$.

Given that **exactly three** of these statements are true, determine the unique value of n.

Ans: 9

7. Given propositions P and Q, if $(P \wedge \neg Q) \vee (\neg P \wedge Q)$ is false and $P \vee Q$ is true, determine the truth values of P and Q.

Ans: Both true (P = True, Q = True).

8. Prove by logical equivalence that $(P \Rightarrow R) \wedge (Q \Rightarrow R)$ is equivalent to $(P \vee Q) \Rightarrow R$.

Ans: Proved.

9. For three propositions P, Q, R, for how many of the 8 possible truth assignments is $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$ true?

Ans: 4.

10. On an island where Knights always tell the truth and Knaves always lie, inhabitant A says: "At least one of us is a Knave." Inhabitant B makes no statement. What are the identities of A and B?

Ans: "A is a Knight and B is a Knave."

Section C Substitution & Structure

1. For non-zero real numbers x, y, which one of the following is sufficient to conclude that $x < y$? *(after TMUA 2016 Paper 2 Q10)*

Ans: E) $x^5 < y^5$.

2. Let P, Q, R, S be four propositions. For how many of the 16 possible truth assignments of (P, Q, R, S) is the compound statement $(P \Rightarrow Q) \Rightarrow (R \Rightarrow S)$ false? *(after TMUA 2019 Paper 2 Q3)*

Ans: C) 3.

3. Let $f(x) = x^2 + bx + c$ with $b, c \in \mathbb{R}$. Consider three statements:

I. $f(x) = 0$ has two distinct real roots.

II. $c < 0$.

III. $b^2 - 4c < 0$.

Which of the following describes the possible numbers of true statements among I, II, III? *(after TMUA 2020 Paper 2 Q12)*

Ans: D) Exactly 0, 1, or 2.

4. Consider propositions P, Q, R. You are given that $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$ is false. Which of the following CANNOT be the truth assignment (P, Q, R)? *(after TMUA 2017 Paper 2 Q7)*

Ans: E) (F, F, F).

5. Five logicians each make a statement, as follows:

Mr P: "Of these five statements, an odd number are true."

Ms Q: "Both statements made by women are true."

Mr R: "My first name is Robert and Mr P's statement is true."

Ms S: "Exactly one statement made by a man is true."

Mr T: "Neither statement made by a woman is true."

How many of the five statements can be simultaneously true? *(after TMUA Specimen Paper 2 Q20)*

Ans: D) 3 only.

6. In the Cartesian plane, let L be a line passing through $(1, 2)$. Let P be the statement: "If the y-intercept of L is negative, then the x-intercept of L is positive." Which of the following must be true? *(after TMUA 2022 Paper 2 Q5)*

Ans: F) I and III only.

7. Propositions P and Q are given. Which of the following statements is logically equivalent to "P is necessary and sufficient for Q"? *(after TMUA 2020 Paper 2 Q20)*

Ans: A) $\neg P \iff \neg Q$.

8. A theorem states: "For all real numbers x, if x is rational, then x^2 is rational." Which of the following three restatements of the theorem are correct? *(after TMUA 2023 Paper 2 Q11)*

Ans: F) II and III only.

Section D Challenge

1. A region is defined by the inequalities $x+y>4$ and $x-y>-2$. Consider the three statements: I. $x>1$ II. $y>2$ III. $(x+y)(x-y)>-12$. Which of the above statements is/are true for every point in the region? *(after TMUA 2016 Paper 2 Q8)*

Ans: B) I only.

2. Prove by contrapositive: "If $n^2 - 2n$ is even, then n is even."

Ans: Proved via the contrapositive "if n is odd, then $n^2 - 2n$ is odd".

3. Prove directly: "If $n > 4$ is such that both $n - 1$ and $n + 1$ are prime, then n is a multiple of 6."

Ans: Proved.

4. Prove directly: "If x and y are positive reals with $x \neq y$, then $\dfrac{x+y}{2} > \sqrt{xy}$."

Ans: Proved.

5. Statement T: "If n is a positive integer such that n^2 is a multiple of 6, then n is a multiple of 6." (a) Is T true? Prove or disprove. (b) Contrast with the fact that "n^2 is a multiple of 4" does *not* force "n is a multiple of 4" (witnessed by $n = 6$). What property of 6, that 4 lacks, makes T true?

Ans: (a) T is true. (b) $6 = 2 \times 3$ is squarefree (a product of distinct primes); $4 = 2^2$ is not.